

# **Task 5: Exploratory Data Analysis (EDA)**

## **Exploratory Data Analysis on the Titanic Dataset**

**Submitted By**

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**Task:** Task 5 – Exploratory Data Analysis (EDA)

# INTRODUCTION

Exploratory Data Analysis (EDA) is an important step in the data analysis process. It involves examining and summarizing datasets using statistical techniques and visualizations to understand the underlying patterns, identify anomalies, detect missing values, and discover relationships among variables. EDA helps analysts gain valuable insights into the data before applying machine learning models or making business decisions.

In this task, Exploratory Data Analysis was performed on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis focused on understanding passenger demographics, survival patterns, feature distributions, and relationships between variables.

## 2. Objective

The primary objective of this task is to perform Exploratory Data Analysis on the Titanic dataset to:

- Understand the structure of the dataset.
- Identify missing values and data quality issues.
- Generate descriptive statistics.
- Analyze relationships between variables.
- Visualize important trends and patterns.
- Summarize meaningful insights from the data.

## 3. Dataset Description

The Titanic dataset contains information about passengers who traveled on the RMS Titanic. The dataset includes demographic details, ticket information, passenger class, fare, and survival status.

### Important Features

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare

- Cabin
- Embarked

## **4. Tools and Technologies Used**

- Google Colab
- Python
- Pandas
- Matplotlib
- Seaborn

## **5. Methodology**

The following steps were performed during Exploratory Data Analysis:

### **Step 1: Import Libraries**

The required Python libraries including Pandas, Matplotlib, and Seaborn were imported.

### **Step 2: Load Dataset**

The Titanic dataset was loaded into a Pandas DataFrame using the `read_csv()` function.

### **Step 3: Data Inspection**

The dataset was examined using:

- `head()`
- `info()`
- `describe()`
- `shape`

to understand the dataset structure and data types.

#### **Step 4: Missing Value Analysis**

Missing values were identified using:

```
df.isnull().sum()
```

This helped identify columns requiring preprocessing.

#### **Step 5: Statistical Summary**

Descriptive statistics such as mean, median, minimum, maximum, and standard deviation were generated using:

```
df.describe()
```

#### **Step 6: Data Visualization**

Several visualizations were created to better understand the dataset.

These included:

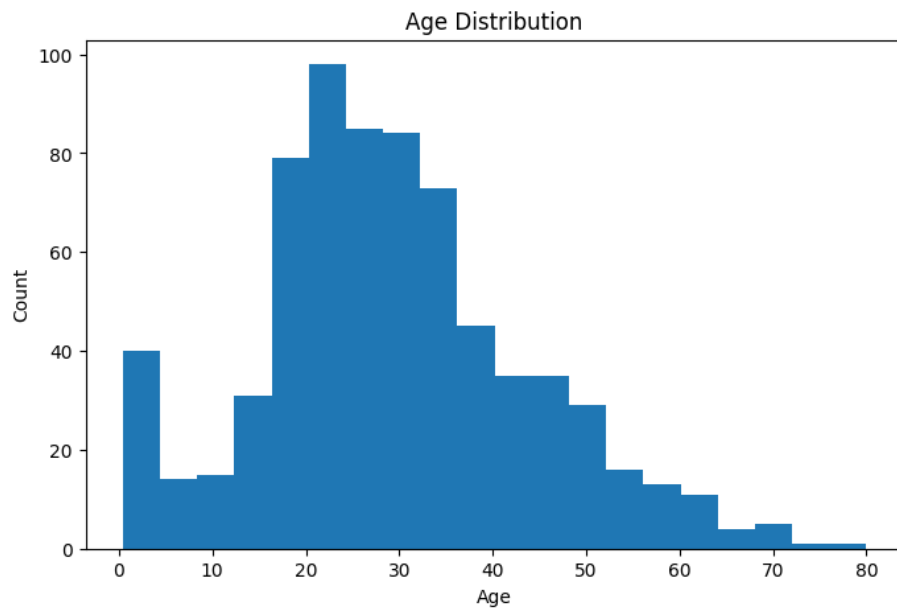
- Histogram
- Boxplot
- Countplot
- Correlation Heatmap
- Pairplot

## **6. Visualizations and Observations**

### **Histogram – Age Distribution**

#### **Observation**

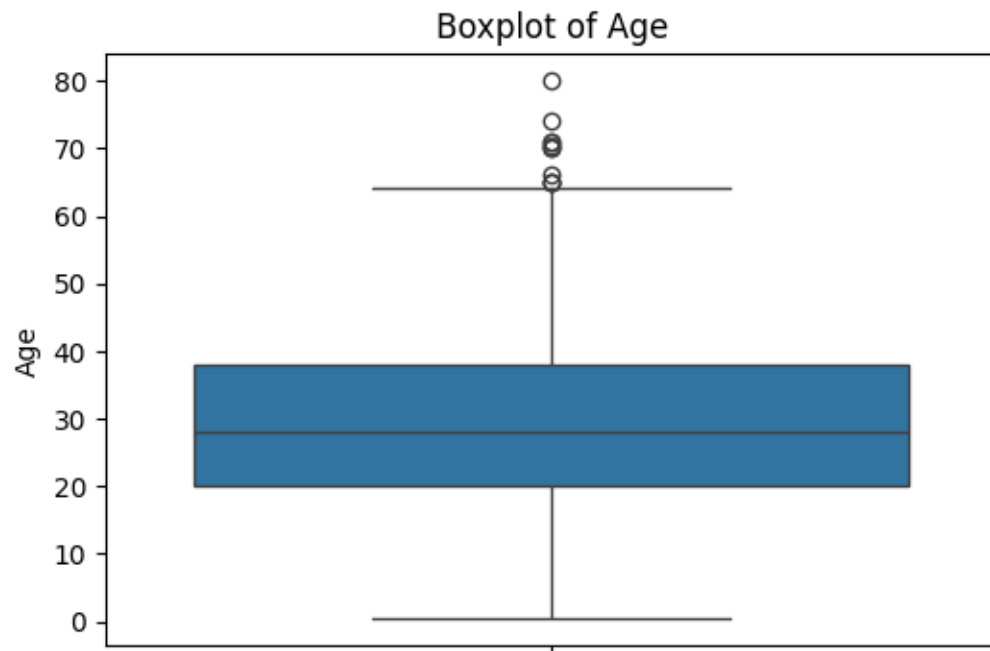
The histogram shows that the majority of passengers were between 20 and 40 years of age. Very few passengers were above 70 years old.



## Boxplot – Age

### Observation

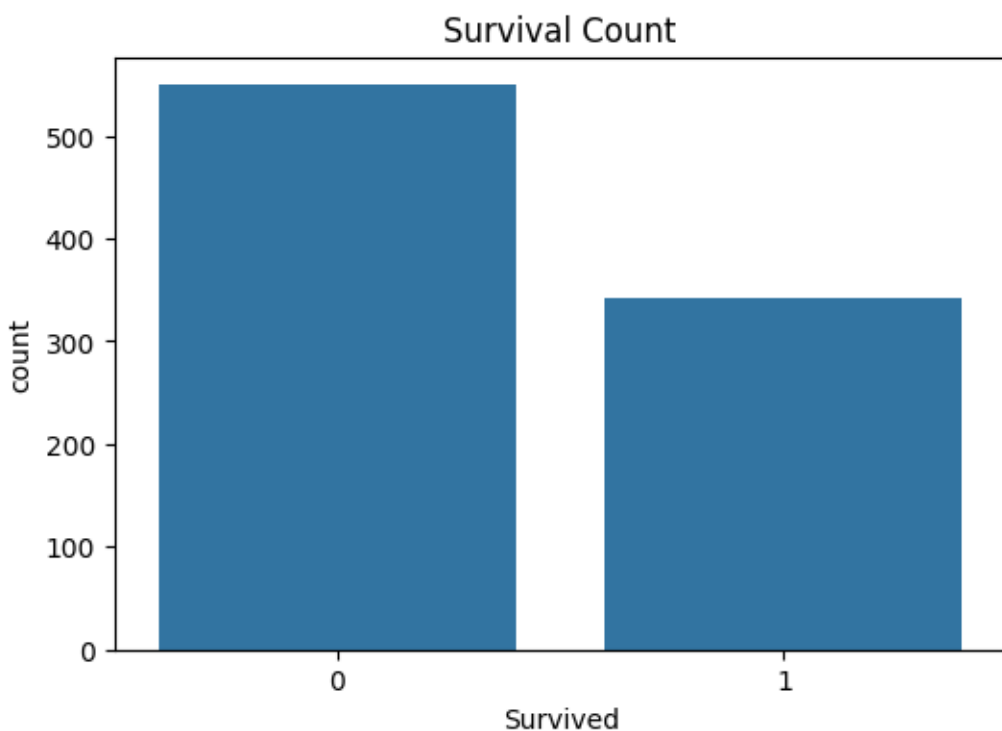
The boxplot reveals several outliers representing elderly passengers. Most passenger ages fall within a moderate range.



## Survival Count

### Observation

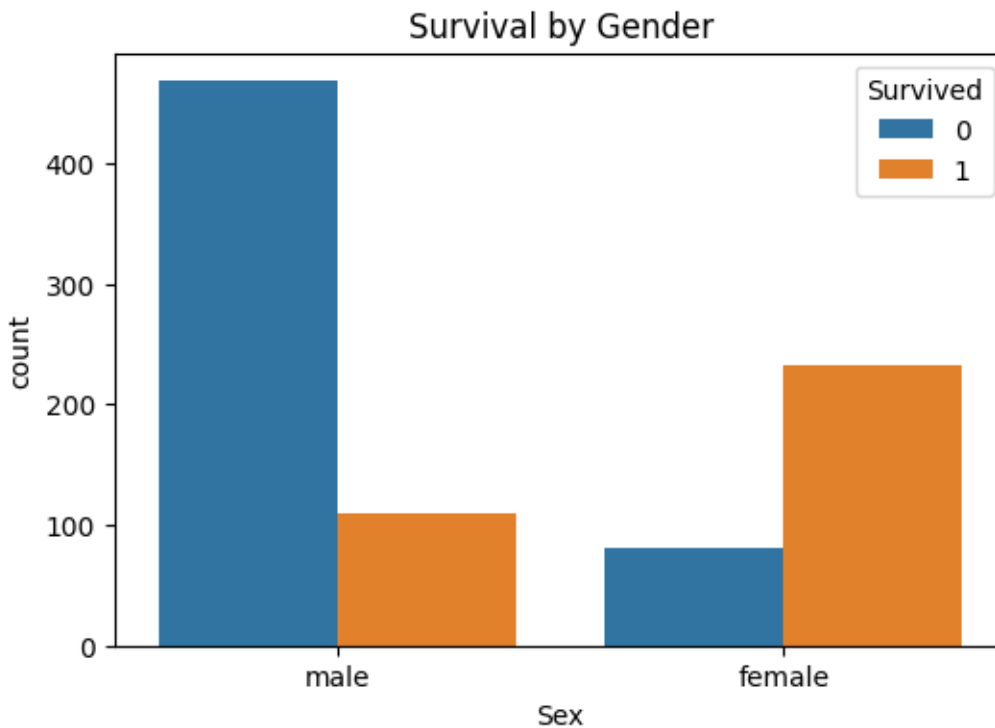
The visualization indicates that more passengers did not survive compared to those who survived the disaster.



## Survival by Gender

### Observation

Female passengers had a considerably higher survival rate than male passengers, indicating that gender influenced survival.

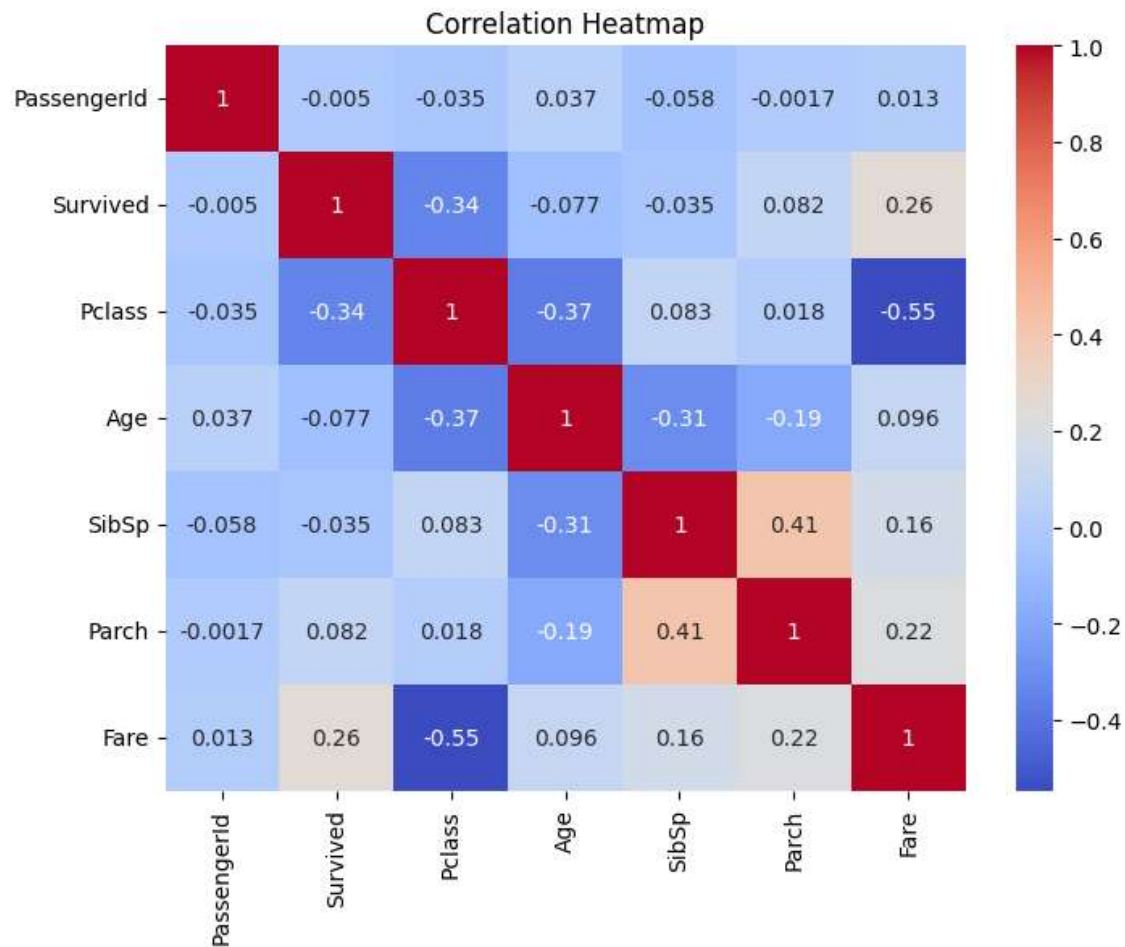


## Correlation Heatmap

### Observation

The heatmap shows relationships among numerical variables. Passenger class and fare exhibit noticeable relationships with survival.



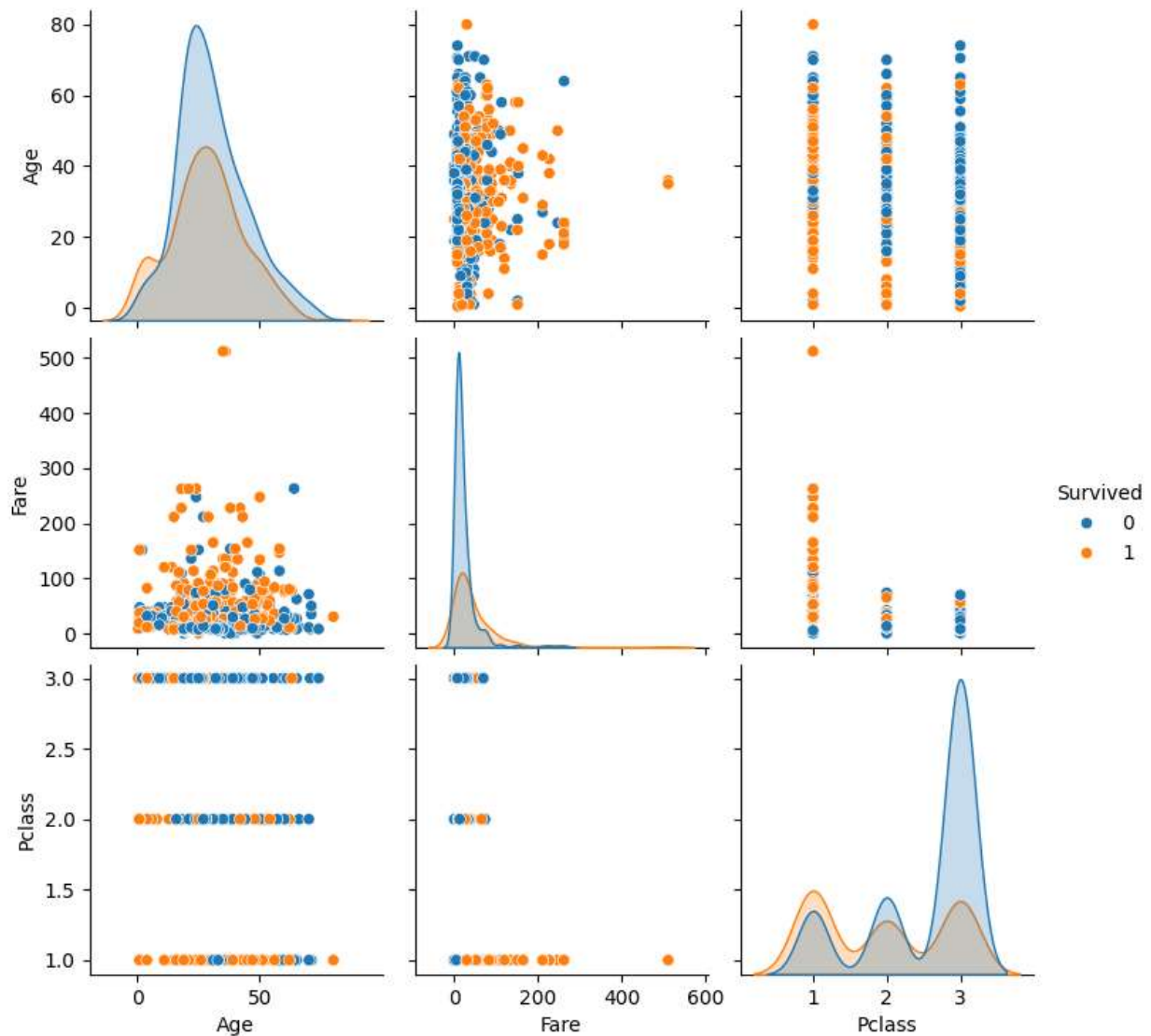


## Pairplot

### Observation

The pairplot illustrates relationships between Age, Fare, Passenger Class, and Survival. It shows that passengers in higher classes with higher fares generally had

better survival rates.



## 7. Key Findings

The Exploratory Data Analysis produced several important insights:

- Most passengers belonged to the 20–40 age group.
- Female passengers had a significantly higher survival rate than male passengers.
- First-class passengers were more likely to survive.
- Fare showed a positive relationship with passenger survival.
- Missing values were present in the Age and Cabin columns.

- Outliers were identified in both Age and Fare.
- Passenger Class had a noticeable impact on survival.

## **8. Conclusion**

The Exploratory Data Analysis successfully provided a comprehensive understanding of the Titanic dataset. Statistical summaries and visualizations revealed important trends, relationships, and data quality issues. The findings demonstrate how EDA helps identify meaningful insights before applying predictive models. This analysis serves as a strong foundation for future data preprocessing and machine learning tasks.